

PROJECT REPORT

Auto Ticket Classification using Flow Designer

ServiceNow Developer Instance

Team ID: SWTID-2026-1711

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Maximum Marks: 10

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1. INTRODUCTION

1.1 Project Overview

Auto Ticket Classification using Flow Designer automates the classification and assignment of ServiceNow service tickets. Based on the configured conditions and the ticket information, the system identifies the appropriate ticket category and automatically routes the ticket to the relevant support group, without manual triage.

1.2 Purpose

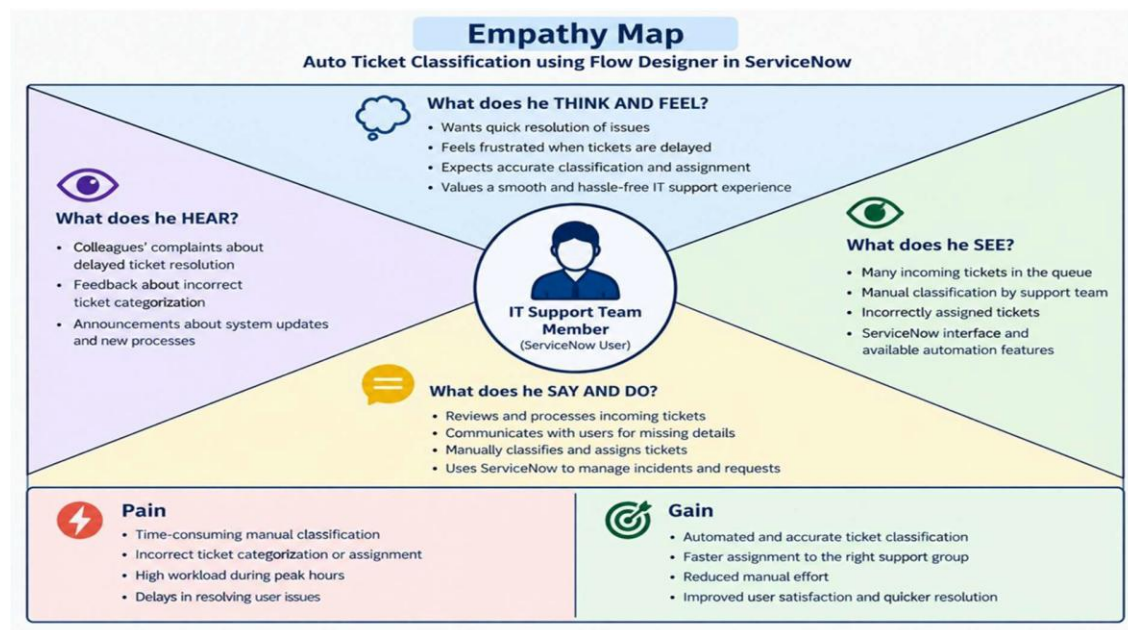
The purpose of this project is to reduce the manual effort and delay involved in sorting incoming tickets, to improve the consistency of classification, and to ensure that every ticket reaches the right support team quickly.

2. IDEATION PHASE

2.1 Problem Statement

Support teams receive tickets that describe hardware, software and network issues. Classifying and assigning each ticket by hand is slow and error-prone, which delays resolution and can send tickets to the wrong team. A rule-based automated workflow is needed to classify tickets and assign them as soon as they are created.

2.2 Empathy Map Canvas



2.3 Brainstorming



3. REQUIREMENT ANALYSIS

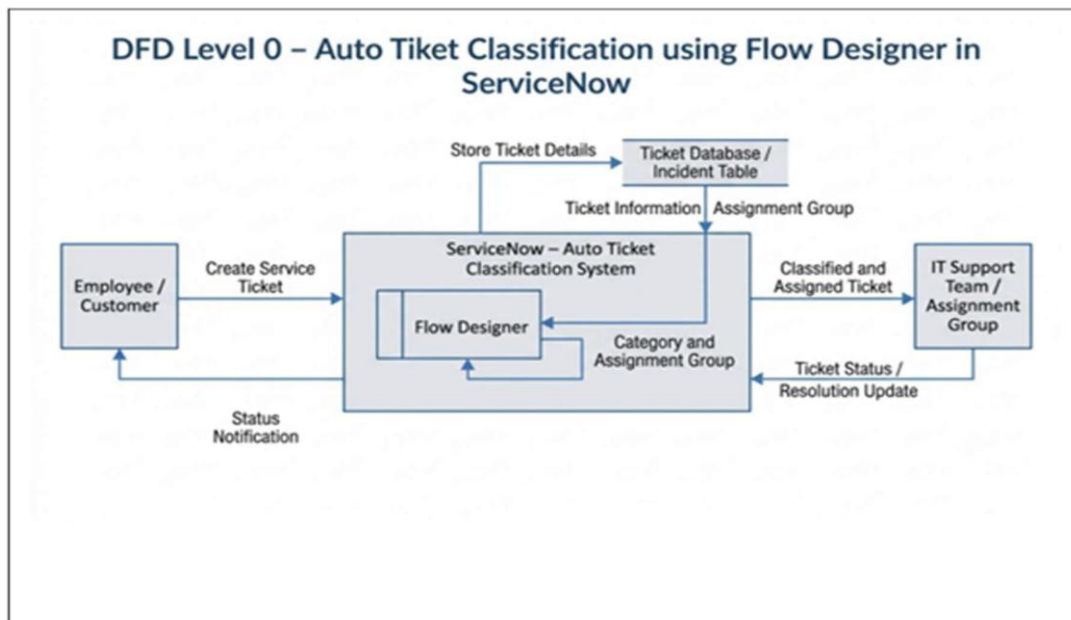
3.1 Customer Journey Map

	1. Create Ticket	2. Enter Details	3. Submit Ticket	4. Auto Classification	5. Auto Assignment	6. Support Processing	7. Resolution & Closure
Steps What does the person typically experience?	Open ServiceNow and create ticket	Enter issue description and details	Submit the incident	Flow Designer identifies ticket category	Ticket routed to correct support group	Support team works on issue	Issue resolved and ticket closed
Interactions What interactions do they have at each step along the way?	ServiceNow Web UI	Ticket form	Incident record	Flow Designer	Assignment group	IT Support Team	ServiceNow status/update
Goals & motivations At each step, what is a person's primary goal or motivation?	"Help me raise my issue easily."	"Help me explain my problem clearly."	"Help me submit my request quickly."	"Classify my ticket automatically."	"Send my ticket to the right team."	"Resolve my issue without unnecessary delay."	"Keep me informed until resolution."
Positive moments What steps does a typical person find enjoyable, productive, or delightful?	Easy ticket creation	Clear issue details	Ticket created quickly	Automatic classification	Correct team receives ticket	Faster processing	Clear resolution status
Negative moments What steps does a typical person find frustrating, confusing, or time-consuming?		Missing/incomplete details		Ticket may not match a rule	Manual assignment may be needed	Processing delays	Delayed closure
Areas of opportunity How might we make each step better?	Improve ticket form	Add clear field guidance	Validate required information	Add/improve classification rules	Improve assignment rules	Automate more actions	Improve notifications and follow-up

3.2 Solution Requirement

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Service Ticket Creation	Create a new service ticket Enter issue description Submit ticket with required details
FR-2	Ticket Configuration	Define ticket categories Configure ticket fields Define support/assignment groups
FR-3	Flow Designer Configuration	Create automated flow Configure flow trigger Configure conditions and flow actions
FR-4	Automatic Ticket Classification	Analyze ticket information Identify appropriate ticket category Automatically update the ticket category
FR-5	Automatic Ticket Assignment	Identify appropriate support group Assign ticket automatically Update assignment group in the ticket
FR-6	Exception Handling	Identify unmatched tickets Keep unmatched tickets for manual review Prevent incorrect automatic assignment
FR-7	Ticket Status & Tracking	Update ticket status Track assigned tickets Allow support team to view ticket details
FR-8	Testing & Validation	Test different ticket types Verify classification results Verify assignment results

3.3 Data Flow Diagram



3.4 Technology Stack

Component	Technology
Platform	ServiceNow Developer Instance
Automation	Flow Designer
Module	Service Management
Environment	ServiceNow Development Environment

4. PROJECT DESIGN

4.1 Problem Solution Fit

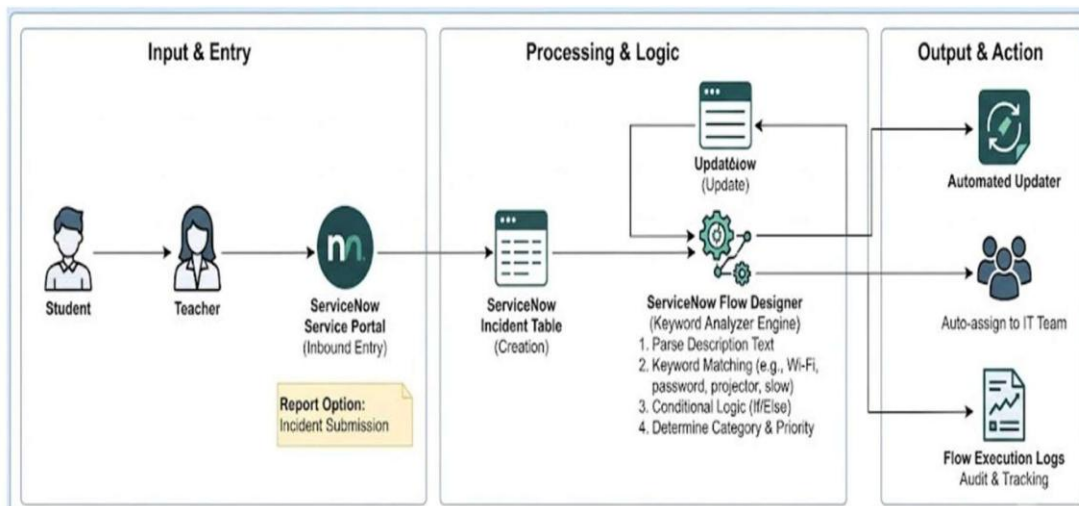
S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The school IT helpdesk receives a high daily volume of incident requests (such as Wi-Fi issues, projector failures, password resets, and slow computers). Manually reviewing each request and assigning categories is time-consuming, inefficient, and creates workflow bottlenecks.
2.	Idea / Solution description	An automated ticket classification system built using ServiceNow Flow Designer. It analyzes keywords within the incident's short description and description fields to automatically assign the correct priority and category to incoming tickets.
3.	Novelty / Uniqueness	Streamlines traditional manual sorting by leveraging built-in ServiceNow Flow Designer logic and keyword analysis, transforming a routine administrative bottleneck into an instant, automated triage workflow without requiring complex, custom-coded external AI models
4.	Social Impact / Customer Satisfaction	Eliminates long resolution wait times for students and teachers, drastically reducing IT staff stress and frustration. It transforms user and staff experience from feeling overwhelmed to feeling confident and relieved through an efficient, reliable workflow.
5.	Business Model (Revenue Model)	Internal operational efficiency model (Internal IT process optimization for educational institutions), which reduces labor costs, lowers administrative overhead, and maximizes resource utilization for school IT departments.
6.	Scalability of the Solution	Highly scalable across any school district or enterprise IT helpdesk environment using ServiceNow. As ticket volumes grow, the Flow Designer logic effortlessly handles increased request loads without requiring a proportional increase in manual triage staff.

4.2 Proposed Solution

A Flow Designer flow is triggered when a ticket is created. The flow evaluates the configured conditions against the ticket information, sets the category, and assigns the ticket to the matching support group. Tickets that match no condition are handled through the configured default logic so that the flow never fails.

1. CUSTOMER SEGMENT(S) CS School IT Helpdesk Staff: Administrators and Technicians	6. CUSTOMER CONSTRAINTS CC Manual categorization is time-consuming, error-prone, and causes delays in incident resolution, impacting service quality.	5. AVAILABLE SOLUTIONS AS Manual review (current, inefficient), potential for custom AI coding (high cost/Complexity).
2. JOBS-TO-BE-DONE / PROBLEMS J&P Accurately categorize a high volume of daily incident tickets (Wi-Fi, projector, password, slow computer issues) to ensure proper assignment and timely resolution.	9. PROBLEM ROOT CAUSE RC The increasing number of incoming requests exceeds the capacity for manual review and accurate classification by IT staff.	7. BEHAVIOUR BE IT staff must read, interpret, and manually assign categories to each incident ticket, causing significant workflow bottlenecks.
3. TRIGGERS TR A notification of a new incident ticket submission in the ServiceNow system. 4. EMOTIONS: BEFORE / AFTER EM Before: Frustrated & overwhelmed. After: Confident & relieved by an efficient, reliable workflow.	10. YOUR SOLUTION SL ServiceNow Flow Designer automation that uses keyword analysis (e.g., 'Wi-Fi', 'projector', 'slow', 'password') within short descriptions to auto-assign a priority and category to incoming tickets.	8. CHANNELS OF BEHAVIOUR CH 8.1 ONLINE: ServiceNow Platform (Service Portal, Flow Designer). 8.2 OFFLINE: N/A (Process is digital).

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint	Task	Duration	Status
Sprint-1	18	6 Days	PASS
Sprint-2	21	6 Days	PASS

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Five functional and three performance test cases were executed. All 8 passed with no failures (100% pass rate).

Test ID	Scenario	Expected Result	Actual Result	Result
FT-01	Ticket Creation Validation	Valid tickets created; validation shown for missing mandatory fields.	Ticket creation and field validation worked as expected.	PASS
FT-02	Automatic Ticket Classification	Ticket classified automatically per configured conditions.	Tickets were classified according to the configured rules.	PASS
FT-03	Automatic Assignment Group Selection	Ticket assigned to the appropriate support group.	Tickets were assigned to the appropriate support groups.	PASS
FT-04	Flow Execution Validation	Flow triggers automatically and executes all actions.	Flow Designer flow triggered and completed successfully.	PASS
FT-05	Unclassified Ticket Handling	Unmatched ticket handled by default logic without flow failure.	Unmatched tickets handled without flow failure.	PASS
PT-01	Flow Execution Time Test	Flow executes within the defined performance target.	Execution completed within the expected time.	PASS
PT-02	Multiple Ticket Processing Test	All tickets processed correctly without failures.	Multiple tickets processed successfully.	PASS
PT-03	Flow Reliability Test	Flow executes consistently for different ticket types.	Flow executed successfully and consistently.	PASS

Testing Type	Test Cases	Passed	Failed
Functional Testing	5	5	0
Performance Testing	3	3	0
Total	8	8	0

7. RESULTS

7.1 Output Screenshots

The screenshot displays the ServiceNow Email configuration page. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Admin', and a search bar. The main header shows 'Email - Your Request for the issue has been submitted.' with a search icon and action buttons (Update, Delete). The configuration form includes the following fields:

- Type: send-ready
- Target: Incident Workflow: INC0001006
- User: (empty)
- Notification type: SMTP
- UID: (empty)
- Created: 2025-09-29 23:01:36
- Deleted: ☐
- Weight: 0
- Importance: (empty)
- Subject: Your Request for the Issue has been submitted.
- Recipients: abel.luter@example.com
- Body:

```
<html><head></head><body><p><p><p data-tnymccootblock=""><strong id="docs internal guid 4a05528c 7f1f c150 7c67 db6865c443be">Ticket confirmation message.</strong></p><p></p></div><div style="display:inline">RefMSG0000149: b1RkV000CtbulFoesk</div></body></html>
```
- Content type: (empty)
- Headers:

```
Message-ID:934432203.0.1790749276820@osp130021.us191.servicenow.com>
X-ServiceNow-Source:FlowDesigner-df7286c7f7684310bd38030c3cfd8e
X-ServiceNow-GeneratedFrom:
X-ServiceNow-SmEmail-Version2
```

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- Removes manual triage and speeds up ticket routing.
- Applies the same classification rules to every ticket.
- Built on Flow Designer, so rules are easy to maintain without heavy coding.
- Unmatched tickets are handled without causing flow failure.

8.2 Disadvantages

- Classification depends on predefined conditions, so new issue types need new rules.
- It does not learn from data, as no machine-learning model is used.
- Unusual or ambiguous descriptions may fall into the default handling.

9. CONCLUSION

9.1 Project Summary

The project automates ticket classification and assignment in ServiceNow using Flow Designer. All functional and performance tests passed, confirming that tickets are classified and routed correctly.

9.2 Key Outcomes

- Tickets are classified automatically according to the configured conditions.
- Tickets are assigned to the appropriate support group.
- The flow handled unmatched tickets and multiple tickets reliably.
- All 8 test cases passed (100%).

10. FUTURE SCOPE

10.1 Enhancement Opportunities

- Add more categories and finer-grained classification conditions.
- Add notifications to assignees and requesters.
- Add dashboards to track classification and assignment performance.

10.2 Future Automation

- Use machine learning or NLP to classify tickets from their descriptions.
- Automate priority assignment and escalation.
- Extend the flow to other ticket types and modules.

11. APPENDIX

11.1 GitHub

<https://github.com/uchihatanziro-cell/Auto-Ticket-Classification-using-Flow-Designer---ServiceNow.git>

11.2 Project Demo Link

<https://drive.google.com/drive/folders/1k74T4fTFXg1D21zHFwUSWsYuT8qtD4nV?usp=sharing>